Effects on the Behavior of Infant Common Marmosets *(Callithrix jacchus)* of Separation from Caregivers and of Drug-Induced Reduction in Caregiver Responsiveness

N.R. CHALMERS
J. LOCKE-HAYDON
Biology Department, The Open University
Milton Keynes, England

The effects of separating 4-week-old twin marmosets for 8 days from their families are compared with the effects of leaving 4-week-old infants with their families, but reducing the responsiveness of their caregivers for 8 days by administration of the tranquilizer fluphenazine decanoate. Both treatments reduce infant mobility, abolish play, and increase the time that infants spend with each other. During the period after the termination of the treatment, both treatment groups continue to play and move less than controls, and try to climb onto the mothers more frequently. These results are consistent with the hypothesis that the principal factor precipitating disturbed behaviors in infant monkeys subjected to long-term separation is disruption of the infant–caregiver relationship rather than the separation itself. Some post-treatment differences occur between separated and tranquilized groups. Notably, reunion after separation appears to involve a marked readjustment of the behavior of all family members to the return of the infants, whereas withdrawal of the tranquilizer does not.

If infant monkeys are separated from their mothers, they become at first very active and vocal. Later they move less and tend to sit in a hunched and immobile posture. These two stages of reaction to separation have been labeled the "protest" and "depression" or "despair" phases respectively (reviewed in Chalmers, 1979; Mineka & Suomi, 1978; Suomi, 1982). Separation disturbs the infant's relationship with its mother by removing the infant physically from her. Attachment theory, as developed by Ainsworth (1979), Bowlby (1969), and Rutter (1979), states that the interference with attachment behavior that accompanies and follows a period of separation stems primarily from disturbance of the mother–infant relationship, rather than from the separation itself. There is some evidence to support this view. Some studies suggest that infants that experience strained relationships with their mothers, or with other principal caregivers, without being separated from them can none the less suffer disturbances to their normal behavior. For example, Egeland and Sroufe (1982), Gaensbauer and Harmon (1982), George and Main

Correspondence to Dr. N.R. Chalmers, Faculty of Science, The Open University, Milton Keynes, MK7 6AA, England
Developmental Psychobiology, 19(5):399–411 (1986)
© 1986 by John Wiley & Sons, Inc.
CCC 0012-1630/86/050399-13$04.00

Received for publication 17 May 1985
Revised for publication 30 September 1985

(1979), and Schneider-Rosen and Cichetti (1984) report that continual maltreatment of human infants affects the security of their attachment to their mothers. Working with rhesus monkeys *(Macaca mulatta),* Hinde (1977) and his coworkers found that infants that were reunited with their mothers after a 13-day separation showed greater distress on reunion when their mothers were not able to give their undivided attention to their infants than when they were able to do so. Other studies show that infants separated from their principal caregivers but given the opportunity to develop relationships with new caregivers suffer less intense behavioral disturbances than do infants who are denied this opportunity. Thus Robertson and Robertson (1971) found that young children who were separated from their parents but who were subsequently cared for in families did not show such severe behavioral disturbance as children of the same ages separated from their parents and admitted to institutions.

If the principal factor responsible for the disturbed behavior of the infant following separation is the disruption of its relationship with its principal caregiver, then we would predict that disturbing the caregiver–infant relationship by experimental manipulations other than separation should nonetheless produce disturbance to infant behavior similar to those shown in response to separation. We report here an experiment on the New World Monkey, *Callithrix jacchus,* the common marmoset, in which responsiveness of the infant's caregivers was experimentally reduced by administration of a tranquilizing drug. The effects on infant behavior of this drug-induced change in responsiveness are compared with the effects produced by separating the infants from their families.

In common marmosets, as in other callitrichids, infants are cared for extensively by fathers and older siblings as well as by mothers (Ingram, 1977; Locke-Haydon & Chalmers, 1983; Stevenson & Rylands, 1985). The drug used in this study was fluphenazine decanoate: a phenothiazine derivative that functions as a long-acting tranquilizer, marketed under the tradename "modecate." Further details of the drug and of its mode of action are given in Locke-Haydon (1983, 1984). Locke-Haydon has shown previously that intramuscular injection of modecate into the fathers of 4-week-old infants reduces the fathers' responsiveness to their infants. Drugged fathers spend less time grooming infants, pick them up less frequently, and approach and leave their infants less often than do control fathers (Locke-Haydon, 1984). In the experiments reported here, not only the fathers, but also the mothers and older siblings of 4-week-old infants received the drug, with the intention of reducing the responsiveness of all of the infant's caregivers.

Methods

Subjects were infant marmosets born at approximately 5-month intervals (mean interbirth interval 159.4 days, sem 3.6 days) into 9 family groups. All but two infants were born as dizygotic twins: the remaining two were singletons. Each family consisted of a mother, father, and two successive sets of twins. When a third set of twins was born, the oldest set of twins, by then aged nearly 11 months, were removed. Family groups were housed in separate cages out of direct sight of each other in three adjacent rooms. Further details of housing and maintenance are given in Chalmers & Locke-Haydon (1981).

The experiment was carried out in three successive phases on infants from nine families: (i) a first control phase carried out on a set of infants from three of the families at a time when they underwent no experimental manipulation: (ii) a treatment phase carried out on a set of infants from all nine families at a time when some of them received

10982302, 1986, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420190502 by Universitaetsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

modecate and others underwent separation; (iii) a second control phase carried out on a set of infants from the nine families at a time when they underwent minimal manipulations to control for the handling experience undergone by treatment phase infants. It was thus possible to compare the behavior performed in both these situations with that shown by controls. It was also possible to compare the behavior of first phase and second phase controls, and hence to investigate whether behavior between infants and caregivers showed long-term changes provoked perhaps by the intervening treatment. Infants who served as subjects in the treatment phase served as older siblings in the second control phase. A comparison of the first and second control phase behavior of older siblings, made it possible to investigate whether the treatment had effects lasting for several months.

Finally, by comparing second-phase control families that during the treatment phase had received modecate, with second-phase controls that during the treatment phase had undergone separation, it was possible to investigate whether differences in the effects of separation and modecate carried over to the succeeding set of twins.

There were 9 first control phase infants. These were reared in families that were totally undisturbed. Their behavioral development was reported in Locke-Haydon & Chalmers (1983). In the treatment phase families were assigned at random either to the drug treatment, in which case they were referred to as "modecate families" or to separation, in which case they were referred to as "separated families." There were 8 separated and 9 modecate infants. Experimental treatments occurred when infants were 4 weeks old and lasted 8 days. In separated families, both infants were removed from their family group, weighed, and housed together in a different room for 8 days in a small (0.46 × 0.48 × 0.6m) cage containing a piece of synthetic, furlike fabric. Hand-feeding of the infants was necessary, the schedule following Stevenson (1976). On the ninth day following separation the infants were returned to their families.

In the modecate group, the parents and older siblings of infants were administered 0.25 mg/kg of fluphenazine decanoate (obtained as modecate from E.R. Squibb & Sons Ltd, and diluted with olive oil) by intramuscular injection when the infants were 4 weeks old. The effects of this drug on caregiving behavior toward infant marmosets have been shown to last for approximately 4 days at this dose (Locke-Haydon 1983). The drug was therefore administered on day 1 and day 5 of the treatment week.

Second control phase infants from families whose previous set of infants had undergone separation were removed at 4 weeks of age, weighed, and then immediately returned to their families. Parents and older siblings from families who had received modecate while rearing their previous sets of infants were injected with an equivalent amount of olive oil vehicle on days 1 and 5 when their second control phase infants were 4 weeks old. There were 13 second control phase infants overall.

Observations were carried out through one-way vision windows by trained observers. Eight 30-minute observation sessions were carried out during each week on treatment and second control phase families at infant ages of 3, 4, 5, 8, and 20 weeks, the father, mother and each infant being the focal animal in two sessions each. For first control phase infants comparable data had been collected at 2, 4, 6, 10, 18, and 22 weeks (Locke-Haydon & Chalmers, 1983). Comparisons between first and second phase controls were thus possible at 4 weeks and at 5 months (20–22 weeks). Behavioral observations were recorded directly onto Apple II microcomputers which were programmed to store information on (i) behavior frequencies and durations; (ii) for behaviors involving social interactions, the identify of the animal who performed the behavior and of the animal at whom it was directed. Behaviors recorded were:

1. Time with. The time a focal animal spent within 20 cm of another animal during the observation sessions.

10982302, 1986, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420190502 by Universitaetsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

2. Time on. The time an infant spent being carried by a caregiver.

3. Rejection. The number of times a caregiver prevented an infant initiating a bout of time on.

4. Tries to get on. The number of successful plus unsuccessful attempts by an infant to initiate bouts of time on.

5. Rub-off attempts. The number of successful plus unsuccessful attempts by a caregiver to end a bout of time on with an infant it was carrying.

6. Cries. The number of distress calls given by the infant. Equivalent to "ngä" (Epple 1968).

7. Grid line cross. A measure of locomotor activity. The frequency with which the focal animal crossed imaginary horizontal and vertical lines dividing the age into four equal compartments. This measure was not available for separated infants during their period of separation since they were housed while separated in substantially smaller cages than during the rest of the experiment.

8. Play. The time spent performing any of the components of social play as described by Chalmers & Locke-Haydon (1981).

9. Approach. Any movement bringing an animal within 20 cm of another.

10. Leave. Any movement taking an animal beyond 20 cm of another.

11. %ap.i − %lv.i. Proximity responsibility index. The percentage of all approaches between an infant and another animal that were due to the infant, minus the percentage of all leavings between them due to the infant. Further details of this index are given by Hinde & Atkinson (1970).

12. Eats. The time spent ingesting and chewing food while stationary.

13. Grooming. The time spend combing through another animal's pelage with hand and teeth.

14. Picks up. The number of times caregivers initiated a bout of time on.

15. Foodstealing. The number of attempts by one animal to take food from another animal. Successful and unsuccessful attempts were distinguished.

16. Sits with. The amount of time that the focal animal spent in stationary physical contact with another animal.

For each observation week, a total frequency and, where appropriate, a total duration was calculated for each social behavior between the infant and each of its caregivers; totals for instances where the infant was the performer and for where it was the recipient of the behavior were calculated separately. (Where required, these totals could subsequently be summed to give, for example, frequency of performance across all caregivers.) Weekly totals were also calculated for the frequency and, where appropriate, the duration of the recorded behaviors that did not involve social interaction. In all, 71 measures of behavior were derived per observation week. For the 2 infants who were singletons, 2 behavior measures were inapplicable and do not enter the analysis; namely, time spent with twin and time playing with twin.

Interobserver reliability was assessed from data collected simultaneously by pairs of observers during half-hour observation sessions. Ten such sessions, selected without regard to infant age or treatment, were undertaken by each pair of observers, and Spearman correlation coefficients were calculated for the 10 pairs of measurements for each pair of observers for each behavior. (For one pair of observers one of whom was involved in only a few observation sessions, only 8 such reliability observation sessions were carried out.) Some behaviors were not performed by infants in the selected sessions, either because they were too old or too young. In all instances where the behaviors were performed in 8 or more of the 10 sessions, correlation coefficients were greater than 0.75.

10982302, 1986, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420190502 by Universitaetsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Results

First Control Phase Versus Second Control Phase

Behaviors of first and second phase controls were compared at 4 weeks and 5 months in order to assess whether a long-term shift in the behavior of control animals had occurred. None of 8 behaviors formally compared at 4 weeks differed significantly (two-tailed t tests, $p < .05$ in all cases.) A further 20 behaviors showed no evidence on inspection of differences in mean values between the two control groups at 4 weeks. Six of 8 behaviors formally compared at 5 months did not differ significantly across the 2 control groups at the 5% level. Second control phase infants at this age, however, approached and left some of their caregivers significantly more often than did first control phase infants. (Infant approaches father, $p < .001$; infant leaves siblings, $p < .002$; two-tailed t tests.) In general, therefore, the behavior of untreated animals did not shift over the duration of the experiment, but approaches and leaves between infants and caregivers at 20–22 weeks, became more frequent. Data on approaches and leaves between infants and caregivers at 20–22 weeks are therefore excluded from subsequent analyses to avoid possible confounding of treatment effects with long-term shifts in behavior unrelated to the treatments. There were no significant differences in the behavior of first and second control phase older siblings. Hence any effects of treatments that the second control phase older siblings had received some 5 months earlier had not persisted throughout the 5-month period.

Post Separation Versus Post Moderate Controls

Behaviors of second control phase families whose previous set of infants had been separated at 4 weeks were compared with behaviors of controls who had received modecate when their previous set of infants had been 4 weeks old. Each behavior was compared across the two groups over post-treatment ages (5, 8, and 20 weeks) using a two-way repeated measures analysis of variance, with age as the repeated measure. No significant main effects were found at the 5% level. The treatment administered to families while rearing one set of offspring did not, therefore, seem to affect the behavior of those families when rearing their next set of offspring.

The lack of differences between post-separation and post-modecate controls and between second and first control phase animals shows that the handling procedures undergone by the second phase controls did not affect their behavior. Thus the effects observed in response to separation and modecate (see below) can be attributed to the separation experience and to the drug itself respectively, rather than to the accompanying handling procedures.

Pretreatment Behaviors in Separated, Modecate, and Second Control Phase Families

44 measures of behavior were compared across the 3 groups when infants were 3 weeks old; that is, during the week before experimental or control treatments were administered. For 43 of the measures there was no significant difference across groups at the 5% level (one way analysis of variance in each case). For one measure, the time that infants spent on the mother, there was a significant difference ($F = 4.07$, df = 2,27, $p < 0.05$). A type I error is not unlikely given the large number of analyses performed. In general, the three groups did not differ in their behavior in the week before treatment.

10982302, 1986, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420190502 by Universitaetsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Behaviors in Separated, Modecate and Second Control Phase Families during Treatment

Since first and second control phase families did not differ in the behaviors discussed here, the second control phase families were selected for comparison with the separated and modecate families. Separation and modecate treatments reduced infant locomotor activity, abolished infant play and increased the amount of time infants spent with their twins, during the treatment week (Fig. 1). Thus, separated infants spent over 99% of their time clinging to one another, immobile. Five out of the 9 modecate infants were totally immobile or made no more than 1 grid line crossing during the week. The other 4 infants were less affected by the treatment. The median number of grid-line crossings for modecate was 1, for second control phase infants 8, ($0.1 > p > .05$, $U = 32$, two-tailed Mann-Whitney U test). Seven of the 13 second control phase infants played during the week, none of the separated or modecate infants did. (Median control values differed significantly from separated and modecate values respectively; $p < .05$, $U = 22$, two-tailed Mann-Whitney U test in both cases.) Since data on the amount of time each infant spent with its twin could be collected for only 1 of each pair of twins, in order to preserve independence of data, sample sizes were too small to allow a formal test across pairs of treatments. However, Figure 1 shows that modecate values for this measure were intermediate between those of second control phase and separated infant values. (One-way analysis of variance across three treatments; $F = 4.81$, df = 2,11, $p < 0.05$).

Comparison between modecate and second control phase families during the period of treatment showed that drugged caregivers, and especially drugged fathers, picked up

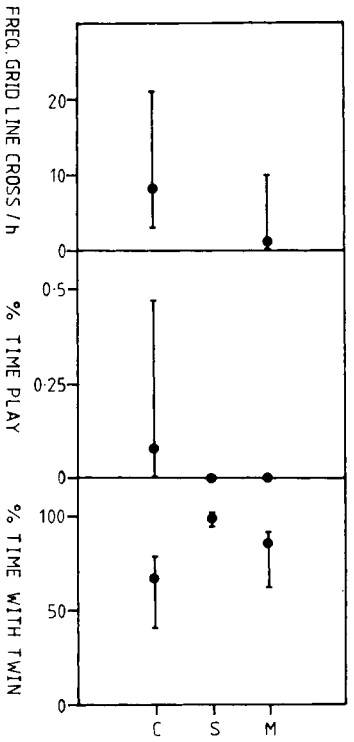


Fig. 1 Locomotor activity as measured by frequency of grid line crosses, percentage of time playing and percentage of time spent with twin by 4-week-old infants. Medians and ranges. C = second control phase infants, S = separated infants, M = infants from modecate families.

10982302, 1986, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420190502 by Universitatsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

infants less than controls and rejected infants less. Fathers receiving modecate also approached infants less, (modecate versus second control phase measures, two-tailed t test, $p < .05$ in all instances, except for frequency of rejections by caregivers, where $p < 0.01$). These effects of modecate on behavior between caregivers and infants are similar to those previously reported with respect to the father (Locke-Haydon, 1983). Infants from modecate families also spent more time on their caregivers than did controls (two-tailed t test, $p < 0.05$). The present results support the conclusion from Locke-Haydon (1983) that modecate reduces the responsiveness of caregivers towards their infants.

Posttreatment Comparisons between Separated, Modecate, and Second Control Phase Families

51 of the 71 behavior measures taken over the post-treatment period showed a significant main effect of age. Age changes in behavior were similar to those previously reported for untreated animals (Locke-Haydon & Chalmers, 1983). Thus infants became increasingly independent of their caregivers, spending less time being carried by them, and less time with each other (Table 1). With this increased independence, infants made fewer attempts to climb onto each of their caregivers, while caregivers rubbed off, rejected and picked up infants less. Infants became increasingly active with age, crossing more grid lines, and playing more. They also cried less and ate more. Foodstealing was the principal behavior not to change significantly with age (Table 1).

Significant treatment main effects included the number of times infants cross grid lines ($F = 6.14$, df = 2,54, $p < .01$), Figure 2; the amount of time spent playing ($F = 6.79$, df = 2,54, $p < .01$), Figure 3; and the number of times infants tried to climb on their mothers ($F = 5.10$, df = 2,54, $p < .05$), Figure 4. The Figures show that separated and modecate infants were less mobile, played less and, with some exceptions (see below) tried to climb on their mothers more than did controls. Behaviors of caregivers toward the infants showed no significant main effects of treatment during the weeks after the treatment ended. Significant age by treatment interactions were of three kinds: (i) Behaviors where modecate and second control phase values showed similar trends with age, even though absolute values for a given age may have differed, whereas separated infants showed a different age trend. Thus separated infants, upon reunion with their families at 5 weeks spent a high proportion of their time on their mothers, and tried frequently to climb onto her and onto other caregivers (Figures 4 and 5). At 8 weeks both of these behaviors decreased substantially, a decrease that was not shown to the same extent by second control phase or modecate infants. (Age × treatment interactions; time on mother: $F = 4.76$, df = 4,54 $p < .01$; no. of times climbs on mother: $F = 4.98$, df = 4,54, $p < .01$; no. of times climbs on all caregivers: $F = 2.81$, df = 4,54, $p < .05$.) The decrease in attempts of formerly separated infants to climb on caregivers at 8 weeks corresponded with a decrease in the frequency of rejection by those caregivers. Caregivers of second control phase and modecate infants, by contrast, showed an increase in their frequency of rejection from 5 to 8 weeks (age by treatment interaction: $F = 2.86$, df = 4,54 $p < .050$. Finally, formerly separated infants show a marked increase in the approach/leave index, from 5 to 8 weeks, whereas control and modecate infants did not (age by treatment interaction: $F = 3.60$, df = 4,54, $p < .05$). Formerly separated infants therefore showed a greater increase in their contribution to the maintenance of proximity with their caregivers than did second control phase or modecate infants. These data therefore show that immediately upon reunion, previously separated infants and their caregivers showed heightened careseeking and caregiving behaviors, followed by a swing in the opposite direction by 8 weeks.

10982302, 1986, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420190502 by Universitatsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. Post-treatment Age Changes in Behaviors. Means, Standard Errors of Means (SE) Plus Summary of Main Effects of Age from Two-Way Repeated Measures Analysis of Variance of Each Behavior.

Behavior	Mean and SE						Age as a main effect
	5 weeks		8 weeks		20 weeks		
	$\bar{x}$	SE	$\bar{x}$	SE	$\bar{x}$	SE	
% time infant on							
father	26.2	± 4.4	3.9	± 1.3	0.3	± 0.2	$F = 28.60$, df = 2,54, $p < .01$
mother	31.0	± 4.4	5.5	± 1.6	0.1	± 0.3	$F = 106.21$, df = 2,54, $p < .01$
siblings	1.4	± 0.4	0.6	± 0.3	0.0	± 0.0	$F = 16.39$, df = 2,54, $p < .01$
all caregivers	58.0	± 5.3	9.8	± 2.0	0.4	± 0.2	$F = 102.04$, df = 2,54, $p < .01$
% time infant with twin	59.1	± 3.1	30.5	± 4.3	23.4	± 2.7	$F = 38.92$, df = 2,22, $p < .01$
No. of times tries to climb on							
father	4.6	± 0.6	2.2	± 0.6	0.5	± 0.2	$F = 44.53$, df = 2,54, $p < .01$
mother	5.4	± 0.5	4.0	± 0.6	0.5	± 0.2	$F = 58.99$, df = 2,54, $p < .01$
siblings	2.2	± 0.4	1.1	± 0.3	0.0	± 0.0	$F = 25.67$, df = 2,54, $p < .01$
all caregivers	12.2	± 1.1	9.7	± 2.5	1.0	± 0.2	$F = 82.15$, df = 2,54, $p < .01$
No. of times infant rubbed off by							
father	7.1	± 1.3	2.5	± 0.9	0.0	± 0.0	$F = 21.33$, df = 2,54, $p < .01$
mother	8.4	± 1.2	1.7	± 0.3	0.1	± 0.1	$F = 72.55$, df = 2,54, $p < .01$
siblings	2.7	± 1.0	0.8	± 0.3	0.0	± 0.0	$F = 14.59$, df = 2,54, $p < .01$
all caregivers	19.3	± 2.3	5.0	± 1.9	0.1	± 0.1	$F = 122.99$, df = 2,54, $p < .01$
No. of times infant rejected by							
father	1.0	± 0.3	0.8	± 0.3	0.2	± 0.1	$F = 4.12$, df = 2,54, $p < .01$
mother	1.5	± 0.3	2.2	± 0.5	0.3	± 0.1	$F = 12.73$, df = 2,54, $p < .01$
siblings	0.7	± 0.2	0.4	± 0.1	0.0	± 0.0	$F = 8.70$, df = 2,54, $p < .01$
all caregivers	3.2	± 0.5	3.5	± 0.7	0.5	± 0.1	$F = 24.44$, df = 2,54, $p < .01$
No. of times infant picked up by							
father	2.1	± 1.1	0.1	± 0.0	0.0	± 0.0	$F = 52.70$, df = 2,54, $p < .01$
mother	0.5	± 0.2	0.0	± 0.0	0.0	± 0.0	$F = 12.63$, df = 2,54, $p < .01$
siblings	0.1	± 0.1	0.0	± 0.0	0.0	± 0.0	$F = 7.37$, df = 2,54, $p < .01$
all caregivers	1.7	± 0.2	0.1	± 0.1	0.0	± 0.0	$F = 111.06$, df = 2,54, $p < .01$
Frequency of grid line crossings	20.3	± 3.7	51.4	± 9.1	133.7	± 18.3	$F = 51.15$, df = 2,54, $p < .01$
% time infant plays	1.1	± 0.3	1.8	± 0.3	20.5	± 4.9	$F = 62.54$, df = 2,54, $p < .01$
No. of times times cried	47.3	± 11.3	63.6	± 27.5	2.3	± 1.4	$F = 18.98$, df = 2,54, $p < .01$
No. of times infant ate solid food	1.8	± 0.4	3.9	± 0.6	8.0	± 1.2	$F = 9.63$, df = 2,54, $p < .01$
No. of times infant tried to steal food	3.0	± 0.7	3.3	± 0.5	2.5	± 0.5	$F = 0.12$, df = 2,54, $p > .05$

(ii) Some effects of modecate were more long-lasting than those of separation. Although separated and modecate infants played less than second phase controls at all post-treatment ages, the effect was by far the most pronounced for modecate infants at 20 weeks (Fig. 3) (age by treatment interaction: $F = 3.48$, df = 4,54 $p < .05$). Also, body weights of modecate infants fell progressively behind those of second phase and separated infants with age (Fig. 6). (Age by treatment interaction: $F = 2.67$, df = 14,133 $p < .01$).

(iii) One effect of modecate was the opposite to that of separation. Whereas the caregivers of formerly separated infants groomed them frequently on reunion, and whereas this grooming decreased markedly by 8 weeks (in line with other age changes shown by

10982302, 1986, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420190502 by Universitatsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

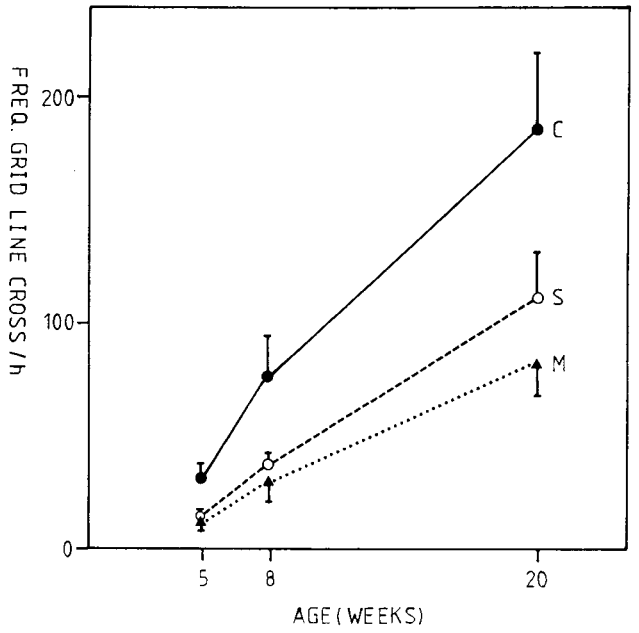


Fig. 2. Locomotor activity as measured by frequency of grid line crosses during post-treatment weeks. Means and s.e.m. C = second control phase infants, S = separated infants, M = infants from modecate families.

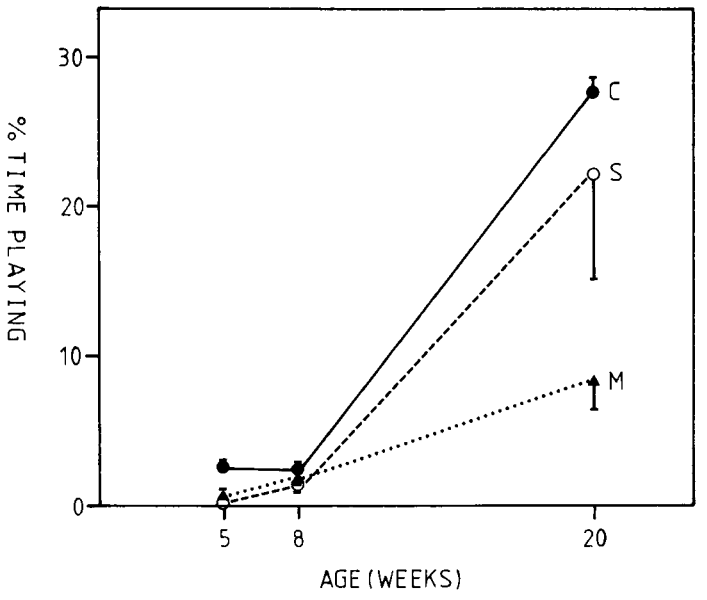


Fig. 3. Mean percentage of time playing during post-treatment weeks. Conventions as in Figure 2.

separated infants listed above), caregivers that had formerly received modecate groomed their infants infrequently at 5 weeks, and increased their frequency of grooming markedly thereafter (Fig. 7), (age by treatment interaction: $F = 3.68$, df = 4,54 $p < .01$).

10982302, 1986, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420190502 by Universitatsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

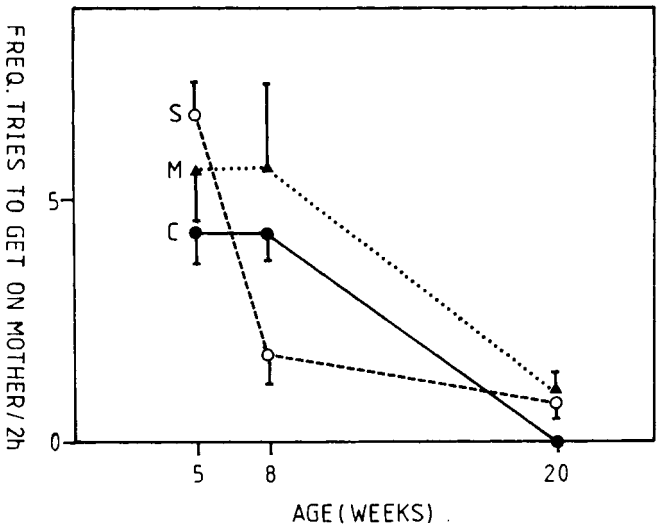


Fig. 4. Mean frequency with which infants tried to climb on to their mothers. Conventions as in Figure 2.

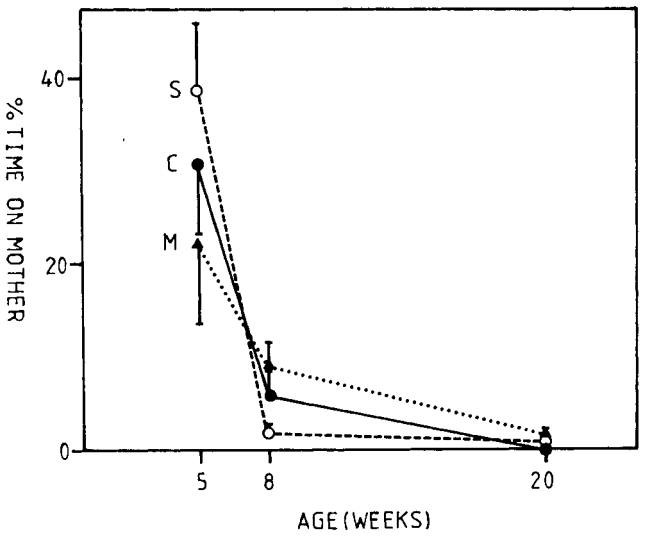


Fig. 5. Mean percentage of time infants spent on their mothers during post-treatment weeks. Conventions as in Figure 2.

Discussion

Our results revealed several similarities between the effects of separation and of reduced responsiveness of caregivers toward their infants. Such similarities were apparent both during the period of experimental treatment when infants were 4 weeks old, and during the weeks following the termination of treatment, at infant ages of 5,8, and 20 weeks. Thus both during and after treatment, separated and modecate infants moved and played less than controls; indeed both treatments totally abolished play in 4-week-olds. During the treatment, separated and modecate infants spent more time with each other than did controls, and during the weeks following treatment they tried to climb onto their mothers more often than did controls.

10982302, 1986, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420190502 by Universitaetsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

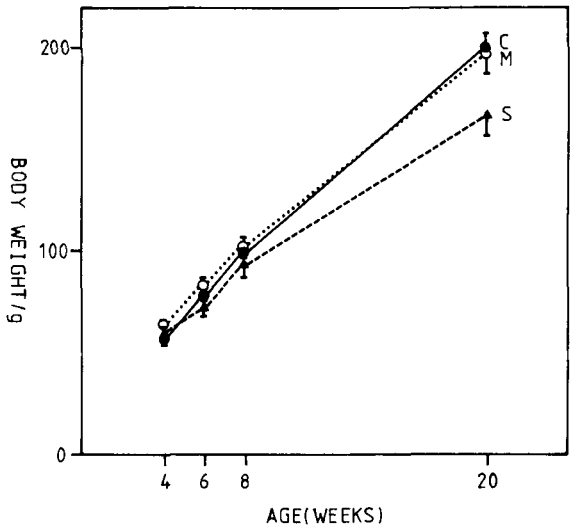


Fig. 6. Mean body weights of infants during treatment and post-treatment weeks. Conventions as in Figure 2.

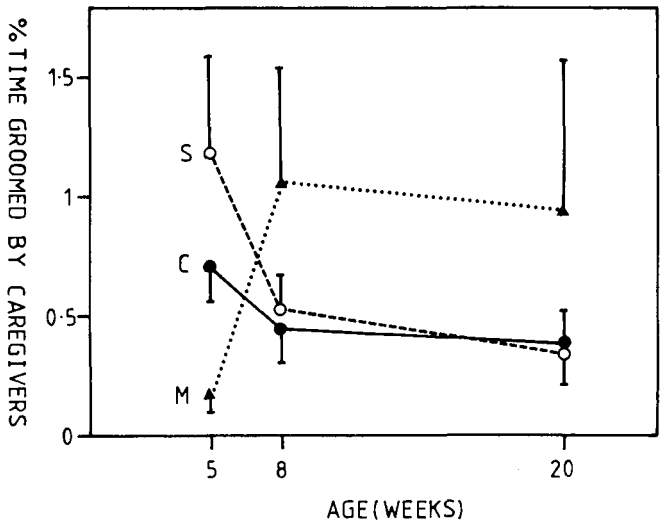


Fig. 7. Mean percentage of time infants were groomed by their caregivers. Conventions as in Figure 2.

It might be argued that the modecate continued to affect caregiver behavior beyond the treatment period and that the post-treatment changes in modecate infant behavior were a direct result of this. This is unlikely to be correct, since the post-treatment behavior of modecate caregivers toward infants was found in general not to differ from that of controls.

The separation effects reported here are similar to those reported for other primate species in the literature, and the effects of modecate reported here confirm those reported earlier by Locke-Haydon (1983). Thus the almost full-time involvement of the infant in huddling with its twin, the decrease in mobility and the abolition of play during the separation period are typical of infant primates of other species raised with their peers while separated from their mothers (Harlow, 1969; Suomi, Harlow, & Domek, 1970).

10982302, 1986, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420190502 by Universitaetsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Furthermore, the relatively long-lasting consequences of the short-term separation reported here are similar to the long-lasting effects of 13 day separations reported by Spencer-Booth & Hinde (1971) in rhesus monkeys. Previously separated rhesus infants tended to be less active and to show less social but more manipulative play than controls.

Our results show that infant behavioral responses to separation from their caregivers are not unique to separation, but also occur in response to reduced caregiver responsiveness. These results are therefore consistent with the hypothesis that disturbance to the caregiver–infant relationship rather than separation itself is primarily responsible for those disturbed behaviors which are seen at their most extreme in infant monkeys subjected to severe and lengthy separation.

An alternative explanation of our modecate results is that the drug passes through the mother's milk to the infant thereby affecting its behavior directly. This explanation is unlikely to be correct for at least two reasons. First, Locke-Haydon (1983) showed that infant marmosets displayed comparable behavioral disturbances to those reported here when only the father received modecate, and hence when the infants cannot have received the drug either in its injected or in its metabolized form through the maternal milk. Second, Locke-Haydon (1983) showed that the effects of modecate on the social behaviors of animals to whom it was administered lasted for no more than 4 days. Since some of the behavioral effects of infants reported here were found 15 weeks after termination of the drug administration, it is unlikely that the drug was acting directly on the infants. Despite these arguments, we are currently carrying out experiments in which modecate is administered to mothers whose lactation has been experimentally blocked, and so we shall be able to test directly whether modecate acts on infants through maternal milk.

It cannot be argued that all of the effects of separation and modecate were identical. Separation and reunion involved changes in family composition, whereas modecate administration did not. The period of reunion following separation involved a period of readjustment both on the part of the infants and on the part of the caregivers. This is clear from the oscillatory changes in behavior of both caregivers and infants that occurred from 5 to 8 weeks, these being very similar to the changes in behavior of mother and infant rhesus monkeys after a 6 day separation (Hinde, 1983; Spencer-Booth and Hinde, 1967). The post-treatment behavior of modecate families showed no such oscillations. Again, the consequences of modecate and separation differed in that the effects on infant weight and play were more severe and long-lasting for infants from modecate than from separated families. We have argued above that such consequences are unlikely to be due to the direct action of the drug through the mother's milk. Reduced responsiveness by caregivers can, therefore, produce some effects that are more serious for development than separation, even though the latter, on the surface, appears to the human investigator to involve the more intrusive experimental intervention.

Notes

This work was supported by a grant from the Medical Research Council of Great Britain. We thank Steven Walters, Sharon Pritchard and Jacqueline Campbell for their skillful care of the animals, Kate Scanlon and Sharon Pritchard for their assistance with the observations and Lynda Birke for her comments on a draft of the paper.

References

Ainsworth, M.D.S. (1979). Attachment as related to mother-infant interaction. *Adv. Study Behav.*, *9*, 2–52.
Bowlby, J. (1969). *Attachment and Loss. Volume 1: Attachment*. New York: Basic Books.
Chalmers, N.R. (1979). *Social Behaviour in Primates*. London: Edward Arnold.

10982302, 1986, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420190502 by Universitaetsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Chalmers, N.R., and Locke-Haydon, J. (1981). Temporal patterns of play bouts in captive common marmosets *(Callithrix jacchus). Anim. Behav., 29,* 1229–1238.

Egeland, B., and Sroufe, L.A. (1981). Attachment and early maltreatment. *Child Dev., 52,* 44–52.

Epple, G. (1968). Comparative studies on vocalizations in marmoset monkeys (Hapalidae). *Folia primatol., 13,* 48–62.

Gaensbauer, T.J., and Harmon, R.J. (1982). Attachment behavior in abused/neglected and premature infants. In R.N. Emde and R.J. Harmon (eds.), *The Development of Attachment and Affiliative Systems.* New York: Plenum. Pp. 263–279.

George, C., and Main, M. (1979). Social interactions of young abused children: approach, avoidance and aggression. *Child Dev., 50,* 306–318.

Harlow, H.F. (1969). Age-mate or peer affectional system. In D. Lehrman, R. Hinde and E. Shaw (eds.), *Advances in the Study of Behavior. Volume 2.* New York: Academic. Pp. 333–383.

Hinde, R.A. (1977). Mother–infant separation and the nature of inter-individual relationships: experiments with rhesus monkeys. *Proc.Roy.Soc.B.Biol.Sci., 196,* 25–50.

Hinde, R.A. (1983). Feedback in the mother-infant relationship. In R.A. Hinde (ed.), *Primate Social Relationships.* Oxford: Blackwell Scientific. Pp. 70–73.

Hinde, R.A., and Atkinson, S. (1970). Assessing the roles of social partners in maintaining mutual proximity, as exemplified by mother–infant relations in rhesus monkeys. *Anim. Behav. 18,* 169–176.

Ingram, J.C. (1977). Interaction between parents and infants and the development of independence in the common marmoset *(Callithrix jacchus). Anim. Behav., 25,* 811–827.

Locke-Haydon, J. (1983). The development of infant-caregiver relationships in the common marmoset *(Callithrix jacchus).* Ph.D. thesis. The Open University.

Locke-Haydon, J. (1984). Injection of a neuroleptic produces changes in caregiver-infant interactions in marmosets. *Behav.Proc., 9,* 261–269.

Locke-Haydon, J., and Chalmers, N.R. (1983). The development of infant-caregiver relationships in captive common marmosets *(Callithrix jacchus). Int. J. Primatol., 4,* 63–81.

Mineka, S., and Suomi, S.J. (1978). Social separation in monkeys. *Psychol. Bull., 85,* 1376–1400.

Robertson, J., and Robertson, J. (1971). Young children in brief separations: a fresh look. *Psychoanal. Study Child, 26,* 264–315.

Rutter, M. (1979). Maternal deprivation, 1972–1978: new findings, new concepts, new approaches. *Child Devel., 50,* 283–305.

Schneider-Rosen, J., and Cicchetti, D. (1984). The relationship between affect and cognition in maltreated infants: quality of attachment and the development of visual self-recognition. *Child Dev., 55,* 648–658.

Spencer-Booth, Y., and Hinde, R.A. (1967). The effects of separating rhesus monkey infants from their mothers for six days. *J.Child Psychol. Psychiat., 7,* 179–197.

Spencer-Booth, Y., and Hinde, R.A. (1971). Effects of brief separations from mothers during infancy on behaviour of rhesus monkeys 6–24 months later. *Child Psychol. Psychiat., 12,* 157–172.

Stevenson, M.F. (1976). Maintenance and breeding of the common marmoset *Callithrix jacchus* with notes on handrearing. *Int.Zoo.Yearb., 16,* 110–117.

Stevenson, M.F., and Rylands, A.B. (1985). The marmoset monkeys, genus *Callithrix.* In A.F. Coimbra-Filho and R.A. Mittermeier (eds.), *Ecology and Behavior of Neotropical Primates. Volume 2.* Rio de Janeiro: Academia Brasileira de Ciências. (In press).

Suomi, S.J. (1982). Abnormal behavior and primate models of psychopathology. In J.L. Fobes and J.E. King (eds.), *Primate Behavior.* New York: Academic. Pp. 171–215.

Suomi, S.J., Harlow, H.F., and Domek, C.J. (1970). Effect of repetitive infant–infant separation of young monkeys. *J. Abnormal Psychol., 76,* 161–172.

Watson, J.S. (1979). Perception of contingency as a determinant of social responsiveness. In E. Thoman (ed.), *Origins of Infants' Social Responsiveness.* New York, Erlbaum.

10982302, 1986, 5, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/dev.420190502 by Universitaetsbibliothek, Wiley Online Library on [01/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License